Exact Itinerary Intervals and Mode Locking for a Contracted Rotation

HCS Research Program

30 August 2026 (revision 0)

Abstract

We give a finite, exact atlas for the discontinuous contracted rotation $f_{\lambda,\delta}(x) = \{\lambda x + \delta\}$ on $[0, 1)$. A binary carry word determines an affine return map with slope λ^n, hence one candidate fixed point. For $\lambda \in \{1/2, 2/3, 3/4\}$ we enumerate all 747 primitive cyclic representatives through length 12 for each slope and intersect the half-open branch inequalities using rational arithmetic. The receipt contains 2241 word rows, 138 nonempty components, exact endpoint equalities, and 295 independent direct-iteration probes. Grouped carry rotation values expose mode-locking components, but we do not call their union maximal. A source-local factor $1 - z^n\lambda^n$ is recorded only as itinerary bookkeeping. There is no arithmetic clock or target determinant; the strict Route-A verdict is `ROUTE_A_REJECTED`.

1 Map, branches, and the question

The fractional-part notation hides a genuine modelling choice at the jump. We freeze the half-open state space $I = [0, 1)$ and write

$$x_{j+1} = \lambda x_j + \delta - k_j, \qquad k_j \le \lambda x_j + \delta < k_j + 1, \qquad k_j \in \{0, 1\}. \tag{1}$$

Thus the lower carry equality belongs to a branch and the upper equality does not. The research question is whether this convention permits an exact finite mode-locking certificate rather than a floating-point picture. The answer below is affirmative at an explicit cutoff, with all global claims carefully withheld.

2 Affine words and exact admissibility

Theorem 1 (Word return formula)**.** *For $w = (k_0, \ldots, k_{n-1})$,*

$$f_w(x) = \lambda^n x + \delta \sum_{r=0}^{n-1} \lambda^r - \sum_{j=0}^{n-1} k_j \lambda^{n-1-j}. \tag{2}$$

Consequently the word has one candidate fixed point

$$x_w(\delta) = \frac{\delta}{1-\lambda} - \frac{K_w}{1-\lambda^n}, \qquad K_w = \sum_{j=0}^{n-1} k_j \lambda^{n-1-j}, \tag{3}$$

and derivative $(f_w)' = \lambda^n$.

Proof. Substitution of (1) gives (2). Since $0 < \lambda^n < 1$, solving $f_w(x) = x$ gives (3), and differentiating (2) gives the derivative. □

Write $x_j(\delta) = a_j\delta + b_j$, beginning with (3), and update $(a_{j+1}, b_{j+1}) = (\lambda a_j + 1, \lambda b_j - k_j)$. The next result is the computational core.

Proposition 1 (Interval equivalence)**.** *The parameter set for which w is an admissible cycle is exactly*

$$\mathcal{D}_w = \{\delta \in [0, 1) : 0 \le a_j\delta + b_j < 1, \quad k_j \le \lambda(a_j\delta + b_j) + \delta < k_j + 1 \ \forall j\}. \tag{4}$$

For rational λ, this is an exactly computable rational interval, possibly empty, with independently recorded endpoint closure.

Proof. The inequalities are necessary by the state domain and (1). If they hold, each selected carry is exactly k_j, so (2) returns the candidate to itself. Every inequality is an affine half-line in δ, and finite intersection preserves exact rational endpoints and their open/closed status. □

Table 1: Exact word-certified components. Every upper endpoint is open under the convention (1). These intervals are not asserted maximal.

λ	word	$\mathcal{D}_w$	$\rho(w)$	λ^n
1/2	0	$[0, 1/2)$	0	1/2
1/2	01	$[2/3, 5/6)$	1/2	1/4
1/2	001	$[4/7, 9/14)$	1/3	1/8
2/3	01	$[3/5, 11/15)$	1/2	4/9
3/4	01	$[4/7, 19/28)$	1/2	9/16

3 Primitive representatives and the finite atlas

A word is primitive when it is not a repetition of a shorter block. We keep the lexicographically least cyclic rotation. Its source-local rotation label is

$$\rho(w) = \frac{1}{n} \sum_{j=0}^{n-1} k_j. \tag{5}$$

The exact factor $1 - z^n \lambda^n$ is useful for checking repetition and derivative bookkeeping, but it is not a target determinant.

The census uses all primitive canonical words of lengths 1 through 12. There are 747 per slope and 2241 rows in total. Only 138 rows have nonempty $\mathcal{D}_w$; grouped rows retain each component interval and its word id. Table 1 gives representative rows (all omitted rows are in the machine-readable receipt).

The endpoint audit evaluates every active equality, including the parameter constraint $\delta < 1$. For example, the lower endpoint 2/3 of the $\lambda = 1/2$, word 01 component satisfies the carry-1 lower equality and is accepted; at 5/6, carry-0 reaches its upper equality and is rejected. This is why a decimal plot alone cannot certify the interval.

4 Independent iteration and literature boundary

For each slope we also probe eight base values $\delta = i/8$ and every distinct endpoint. A separate implementation runs 360 iterations at 90 decimal digits, detects a repeated suffix of length at most 12, and compares its affine fixed point with the exact rational candidate. The 295-row ledger is a control for the finite certificate, not a claim beyond the cutoff.

Laurent and Nogueira study this contracted-rotation family and its rotation number, including algebraic-parameter phenomena [1]. The interval piecewise-contraction results of Nogueira and Pires give a finite periodic-orbit bound under their stated hypotheses [2]. No global one-periodic-orbit theorem is claimed here: their general two-branch theorem gives only an at-most-two bound under its hypotheses. Bugeaud and Conze supply the contracting-mod-one and Farey/Hecke–Mahler context [3]. Our contribution is the reproducible finite Fraction census and boundary ledger, not a priority claim.

References

[1] M. Laurent and A. Nogueira, "Rotation number of contracted rotations," *Journal of Modern Dynamics* 12 (2018), 175–191, DOI: 10.3934/jmd.2018007.

[2] A. Nogueira and B. Pires, "Dynamics of piecewise contractions of the interval," *Ergodic Theory and Dynamical Systems* 35 (2015), 2198–2215, DOI: 10.1017/etds.2014.16.

[3] Y. Bugeaud and J.-P. Conze, "Calcul de la dynamique d'une classe de transformations linéaires contractantes mod 1 et arbre de Farey," *Acta Arithmetica* 88 (1999), 201–218, DOI: 10.4064/aa-88-3-201-218.

Declarations

Scope. `NO_BAD_EULER_OR_ROOT_NUMBER`; all determinant language is source-local itinerary bookkeeping. No target arithmetic, target-zero, Euler-factor, root-data, automorphy, or Hilbert–Pólya claim is made. **Data and code.** Exact rows, checker, replay, and mutation audit are included in the release. This is not external peer review. **AI-use disclosure.** Generative tools assisted drafting and code generation; displayed formulas and metadata are checked by the deterministic artifact chain.